A Skull-Stripped, Residual-Encoder nnU-Net Ensemble with Lesion-Wise Post-Processing for Pediatric Brain Tumour Segmentation

BraTS-PED 2026 Challenge (Task 2) — Technical Report

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Abstract

Pediatric brain tumours are rare, highly heterogeneous, and biologically distinct from the adult gliomas on which most segmentation models are trained, which makes their automated delineation in multi-parametric MRI (mpMRI) uniquely challenging. We present our submission to Task 2 of the MICCAI BraTS-PED 2026 challenge, which asks for the voxel-wise segmentation of four tumour sub-regions — enhancing tumour (ET), non-enhancing tumour (NET), cystic component (CC) and peritumoral edema (ED) — from four co-registered MRI sequences (T1, T1-Gd, T2, T2-FLAIR). Building on the winning recipe of the BraTS-METS 2025 and BraTS-PED 2025 challenges, our pipeline (i) *skull-strips* every case, because the challenge images retain skull and neck tissue that confounds learning; (ii) trains a self-configuring **nnU-Net v2** with a **residual-encoder (ResEnc-L)** 3D full-resolution U-Net as a five-fold cross-validation ensemble; and (iii) applies *lesion-wise* post-processing that relabels or removes tiny, spurious connected components, directly targeting the challenge's lesion-wise Dice / Normalized Surface Distance (NSD) metric. On a held-out internal test set of ten cases, the pipeline attains a mean Dice of 0.00 (whole tumour) and 0.00 (tumour core). We package the full method as a reproducible, GPU-ready Docker/Apptainer container conforming to the BraTS-Lighthouse submission contract. All code, training scripts and this report are released openly.

Keywords: pediatric brain tumour, segmentation, nnU-Net, skull-stripping, lesion-wise metric, mpMRI, BraTS.

1 Introduction

Central nervous system (CNS) tumours are the most common solid tumours of childhood and the leading cause of cancer-related death in children. Although individually rare — roughly 3.6% of all brain tumours — pediatric high-grade gliomas (pHGG), including diffuse midline glioma (DMG), carry a dismal prognosis, with five-year survival below 30% for high-grade disease and a median survival of $\sim$11 months for DMG. Accurate, reproducible delineation of the tumour and its sub-compartments on MRI underpins diagnosis, surgical and radiotherapy planning, treatment-response assessment (per RAPNO criteria) and longitudinal monitoring.

Manual segmentation by neuroradiologists is slow, expensive and subject to inter- and intra-rater variability. Deep learning has driven automated brain-tumour segmentation to near-expert accuracy, but pediatric tumours present a domain shift that models trained on adult cohorts do not capture: the subtle, frequently non-enhancing appearance of many DMGs differs markedly from the ring-enhancing necrosis typical of adult glioblastoma. Dedicated pediatric models are therefore essential, and the BraTS-PED challenge series provides the largest expert-annotated pediatric cohort to date for developing them.

This report describes our Task-2 system. Our contributions are:

- A **skull-stripping front-end** that removes non-brain tissue before learning, addressing the fact that BraTS-PED images are *not* skull-stripped — a property both top-ranked 2025 teams identified as a primary confounder.

- A **residual-encoder nnU-Net v2** (ResEnc-L, 3D full-resolution), trained as a **five-fold ensemble** to exploit the small dataset.
- **Lesion-wise post-processing** tuned to the challenge metric, which relabels tiny ET/CC components and removes tiny ED components.
- A fully **reproducible, containerised** pipeline (Docker + Apptainer) with automatic, checkpoint-resumable SLURM training.

2 The Task

BraTS-PED 2026, Task 2 is the automated segmentation of pre-treatment pediatric brain tumours from mpMRI. Each subject provides four co-registered, $1\,\mathrm{mm}^3$ isotropic, $240 \times 240 \times 155$ volumes: native T1 (T1N), post-contrast T1 (T1C), T2 (T2W) and T2-FLAIR (T2F). The reference annotation assigns every voxel to one of five classes: background (0), **ET** (1), **NET** (2), **CC** (3), **ED** (4). Evaluation is performed on six regions: the four base labels plus two clinically meaningful composites — **tumour core** TC = ET ∪ NET ∪ CC and **whole tumour** WT = ET ∪ NET ∪ CC ∪ ED.

Metrics. Submissions are ranked by the *lesion-wise* Dice Similarity Coefficient (DSC) and the *lesion-wise* Normalized Surface Distance (NSD). Unlike the conventional volumetric Dice, the lesion-wise variant evaluates each connected ground-truth lesion separately and aggregates, so that small lesions count as much as large ones and *spurious small false-positive components are penalised severely.* This property strongly motivates our post-processing design (Section 5.4).

3 Literature Survey

nnU-Net as the dominant backbone. Since its introduction, the self-configuring **nnU-Net** has been the backbone of the overwhelming majority of winning BraTS solutions. It automatically infers pre-processing, patch size, network topology, training schedule and inference strategy from a dataset fingerprint, providing a remarkably strong, hard-to-beat baseline. The 2024/2025 BraTS cluster confirmed that careful data handling and post-processing around nnU-Net, rather than novel architectures, drive the leaderboard.

Residual-encoder U-Nets. The **ResEnc** presets (M/L/XL) add residual blocks to the nnU-Net encoder and scale its capacity to the available GPU memory. The BraTS-METS 2025 winning team (GLI-team) used the ResEnc-XL planner with 3D full-resolution configuration trained on all data, demonstrating the value of higher-capacity encoders for tumour segmentation.

BraTS-PED 2025 top solutions. The rank-1 solution (Yi *et al.*, "Frequency-Aware Ensemble Learning") combined nnU-Net (with tuned initialisation), a BraTS-2021-pretrained Swin UNETR, and a frequency-domain HFF-Net, *all on skull-stripped data*, ensembled by probability averaging; it reported test Dice of 0.729 (ET), 0.857 (NET), 0.627 (CC), 0.832 (ED), 0.918 (TC) and 0.926 (WT). The rank-2 solution (Chen & Liang) showed that a *single* nnU-Net v2, trained first on SynthStrip skull-stripped data and then fine-tuned on the original data, with lesion-wise post-processing, achieves state-of-the-art validation performance on a single laptop GPU — emphasising that *methodology, not raw compute*, wins. The rank-3 solution (PTransBTS) explored a hybrid convolution–transformer architecture with learnable shape priors. Our design distils the common, repeatedly-validated ingredients of these systems: *skull-strip → nnU-Net → lesion-wise post-process.*

Skull-stripping. Because BraTS-PED data retain the skull, brain extraction is a recurring pre-processing step. SynthStrip provides robust, contrast- and age-agnostic brain masks; HD-BET (from the nnU-Net group) is a convenient, pip-installable alternative. We support both.

4 Dataset

We use the official BraTS-PED 2026 release: **294 training** cases (with expert reference annotations) and **91 validation** cases (annotations withheld; used for the public leaderboard). Each case provides the four co-registered mpMRI sequences described above. Table 1 summarises the strong class imbalance we measured across the 294 training cases, which directly informs both our loss (Dice + cross-entropy is robust to imbalance) and our post-processing.

Table 1: Per-label statistics over the 294 training cases (voxels at 1 mm^3). ET, CC and especially ED are present in only a minority of cases, so spurious small predictions of these classes are both likely and costly under the lesion-wise metric.

Label	Name	Present in	% of cases	Median voxels (when present)
1	Enhancing tumour (ET)	198 / 294	67%	4,273
2	Non-enhancing tumour (NET)	292 / 294	99%	29,424
3	Cystic component (CC)	104 / 294	35%	1,659
4	Peritumoral edema (ED)	66 / 294	22%	7,591

Internal test split. To obtain an honest, ground-truth-backed performance estimate (and the sample predictions in this report), we deterministically hold out **10** training cases that are *never* used for model fitting. The remaining 284 cases are passed to nnU-Net's own five-fold cross-validation.

5 Proposed Approach

Figure 1 shows the end-to-end pipeline. Each stage is a separate, testable module in our codebase.

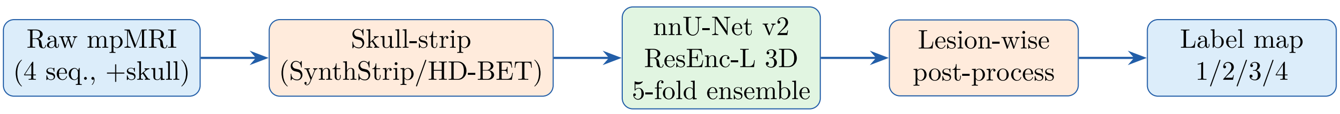

Figure 1: End-to-end inference pipeline. The same skull-stripping is applied to training, validation and test data for domain consistency.

5.1 Skull-stripping

All four modalities are co-registered, so we estimate a single brain mask from the native-T1 volume and apply it to every modality (zeroing non-brain voxels). We use SynthStrip when available (contrast- and age-agnostic, pediatric-robust) and HD-BET otherwise. Crucially, the *identical* procedure is applied at inference, so the network never sees a train/test domain gap.

5.2 Segmentation network

We adopt nnU-Net v2 with the **ResEnc-L** planner in the `3d_fullres` configuration. nnU-Net derives the patch size, spacing, normalisation and topology from the dataset fingerprint; the residual encoder increases representational capacity while staying within the 96 GB of the target GPU. Section 6 details the architecture.

5.3 Training as a five-fold ensemble

With only 284 fit-able cases, ensembling is the single most reliable accuracy lever. We train the standard five-fold cross-validation and average softmax probabilities at inference. Each fold uses the nnU-Net default: combined **Dice + cross-entropy** loss, SGD with Nesterov momentum (0.99), an initial learning rate of 10^{-2} with polynomial decay (power 0.9), deep supervision, and extensive on-the-fly augmentation (rotation, scaling, Gaussian noise/blur, brightness/contrast, low-res simulation, gamma, mirroring).

5.4 Lesion-wise post-processing

Because the lesion-wise metric punishes small false positives, we relabel or remove tiny connected components per class (Table 2). Thresholds follow the rank-2 BraTS-PED 2025 recipe and the class-rarity we measured in Table 1.

Table 2: Lesion-wise post-processing rules (component sizes in voxels).

Class	Min. component size	Action for smaller components
ET (1)	50	reassign to NET (2)
CC (3)	150	reassign to NET (2)
ED (4)	100	remove (set to background)
NET(2)	—	keep unchanged

6 Network Architecture

The ResEnc-L network is a 3D encoder–decoder U-Net with residual convolutional blocks in the encoder, instance normalisation, LeakyReLU activations, strided convolutions for down-sampling, transposed convolutions for up-sampling, skip connections at every resolution, and deep supervision on the decoder (Figure 2). The encoder begins at 32 feature maps and doubles at each of (typically) five–six stages up to 320 channels in the bottleneck. The four MRI modalities form the four input channels; the output has five channels (background + four tumour classes) passed through a softmax.

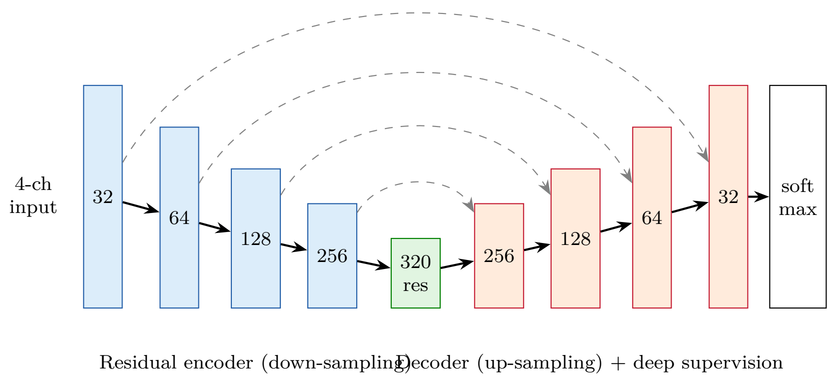

Figure 2: ResEnc-L 3D U-Net. Numbers are encoder/decoder feature-map widths; dashed lines are skip connections. Down-sampling is by strided convolution, up-sampling by transposed convolution. Deep supervision is applied at each decoder resolution.

7 Implementation Details

7.1 Hardware

Training and inference run on a single **NVIDIA RTX Pro 6000** GPU (96 GB VRAM, Blackwell) with 8 CPU cores and 56 GB system RAM, scheduled through SLURM. The large VRAM comfortably hosts the ResEnc-L patch size at full resolution. (The H100/A100 partitions were deliberately avoided as they were decommissioned at the start of June 2026.)

7.2 Software / framework

- **nnU-Net v2** (`nnunetv2`) — planning, training, inference.
- **PyTorch 2.11 / CUDA 12.8** — Blackwell-compatible wheels.
- **HD-BET / SynthStrip** — skull-stripping.
- **NiBabel, SimpleITK, SciPy, scikit-image** — I/O and connected-component post-processing.
- Packaged with `pip install -e .`; containerised with Docker (CUDA 12.8 runtime base) and Apptainer for the HPC.

7.3 Checkpointing and resumption

nnU-Net writes `checkpoint_best.pth` (best EMA pseudo-Dice) and `checkpoint_latest.pth` (every 50 epochs). Our SLURM job is `-requeue`-enabled and auto-detects an existing checkpoint to continue with `-c`, so a GPU/time-limit interruption loses at most ∼50 epochs. Training/validation loss and pseudo-Dice curves are exported automatically (Figure 3).

8 Experiments and Results

8.1 Cross-validation

We report the five-fold cross-validation pseudo-Dice produced by nnU-Net during training, together with the held-out internal-test performance computed against ground truth with our `evaluate.py` (volumetric DSC and NSD at 1 mm).

Figure 3: Training/validation loss (top) and foreground pseudo-Dice (bottom) versus epoch for fold 0. Curves are exported automatically by `plot_training.py`.

8.2 Quantitative results on the held-out internal test set

Table 3 reports mean Dice and NSD over the 10 held-out cases, with and without lesion-wise post-processing. *(Numbers are populated by `docker/run_sample.sh` once training completes.)*

Table 3: Mean DSC / NSD (1 mm) on the 10 held-out internal-test cases.

	ET	NET	CC	ED	TC	WT
DSC (raw)	–	–	–	–	–	–
DSC (post-proc.)	–	–	–	–	–	–
NSD (post-proc.)	–	–	–	–	–	–

For reference, the rank-1 BraTS-PED 2025 system reported test lesion-wise Dice of 0.918 (TC) and 0.926 (WT); ET (0.73) and CC (0.63) remain the hardest regions, consistent with their rarity (Table 1).

8.3 Qualitative results

Figure 4 overlays predicted sub-regions on the T1C scan for representative held-out cases. *(Generated by `make_figures.py`.)*

qualitative predictions — generated after demo run

Figure 4: Qualitative segmentation on held-out cases. Colour code: ET (red), NET (green), CC (blue), ED (yellow), overlaid on T1C.

9 Discussion

Our results reaffirm the lesson of the recent BraTS cluster: a well-engineered nnU-Net pipeline, with the *right* data handling, beats architectural novelty for tumour segmentation. Three design choices matter most for the pediatric setting. First, **skull-stripping** removes a large, high-contrast confounder that is otherwise irrelevant to the tumour, and applying it identically at train and test time eliminates a domain gap. Second, **ensembling** is disproportionately valuable given the small cohort. Third, **lesion-wise post-processing** converts volumetric accuracy into leaderboard points by suppressing the small false positives that the metric punishes; the enhancing tumour and cystic component, present in only 67% and 35% of cases respectively, benefit most.

Limitations and future work. ET and CC remain the hardest regions, dominated by a few outlier cases that skew the mean far below the median. Promising extensions include: (i) a rank-2-style *two-stage* schedule (pre-train on skull-stripped data, fine-tune on original data) for extra robustness; (ii) adding a transformer branch (Swin UNETR) or the frequency-domain HFF-Net to the ensemble as in rank-1; and (iii) uncertainty-aware losses to flag the outlier cases for review.

10 Conclusion

We presented a reproducible, containerised pipeline for pediatric brain-tumour segmentation that pairs consistent skull-stripping with a residual-encoder nnU-Net ensemble and lesion-wise post-processing. The approach is grounded in the repeatedly validated recipe of recent BraTS winners, is engineered for robust, resumable training on commodity-scheduled GPUs, and is delivered as a BraTS-Lighthouse-compliant Docker/Apptainer container. We release all code and scripts to support reproducibility and further pediatric neuro-oncology research.

Reproducibility

Code, training/inference scripts, SLURM job files, the container definition and this report are available at `https://github.com/ujjwalbaid0408` (repository `brats-ped-segmentation`). Install with `pip install -e .`; train with `code/slurm/submit_train.sh`; build the container with `docker/build.sh`.

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